

Case study: Strategy Business Administration Interactive XR Learning

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ABSTRACT

Adopting novel approaches in business administration teachings is difficult as conventions are based on case study readings. We evaluated the use of immersive virtual reality (VR) case studies for business administration strategy studies. The use of VR case studies was well received by the students and improved their understanding. This approach can revolutionize business administration teachings and serve as a model for other fields looking to incorporate innovative education methods.

Keywords: Immersive learning, Interactive learning. Strategy education.

Index Terms: Human-centered computing ~Human computer interaction (HCI) ~HCI design and evaluation methods ~Usability testing.

1 INTRODUCTION

Immersive learning is a type of experiential training that utilizes virtual reality simulations or 360-degree video recordings to create engaging and interactive learning experiences. Studies have shown that immersive learning can lead to higher levels of engagement and improved learning outcomes compared to traditional methods [1]. Additionally, research suggests that 360-degree video can facilitate an interaction between memory and perception that leads to enhanced cognitive processes [2], [3]. Immersive learning has been widely used for a variety of applications, including workplace training [4], medical team training[5], sexual harassment Resilience training [6] and more. In recent years, the use of immersive technologies has grown in popularity, mainly in the education [7] and medical [8] fields.

As the COVID-19 pandemic led to widespread quarantine measures, including the closure of schools and universities, we focused on developing virtual case study scenarios as a means of continuing to deliver business education to students. Harvard-style case studies have long been a popular method for teaching business concepts in undergraduate and MBA programs, but students have grown increasingly disinterested in reading lengthy written case studies. In response, we propose the implementation of immersive virtual reality experiences as a new form of case study that allows students to virtually become part of a business experience and fully immerse themselves in the protagonist's workplace and the specific business dilemma being addressed. This approach has the potential to increase student engagement and improve learning outcomes.

The faculty at our institution integrated the use of 360-degree video recordings and immersive storylines into the classroom as a means of teaching business strategy and marketing. These immersive videos were integrated into dedicated class structures and served as an alternative to traditional case studies. This innovative approach allowed students to experience case studies more engagingly and interactively rather than simply reading them. In addition to improving the use of student time, this approach has been shown to enhance knowledge retention, involvement, and attention[9]. By being able to experience case studies firsthand and collaborate with their professors to guide the direction of the storyline, students are more fully immersed and engaged in the learning process. The use

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of immersive video recordings have proven to be a valuable tool for improving the effectiveness of business education (Figure 1).



Figure 1: Immersive learning with Professor Niron Hashai (A) at Reichman University, students experience the virtual scenario in class (B) outside (C) and at their homes (D).

2 THE STORY

Gilad, Bar and Dror are the owners of the Imperial Bar, a popular establishment located in the Tel Aviv Imperial Hotel. In response to the COVID-19 pandemic and the resulting restrictions on businesses, the management team at the Imperial Bar were faced with the challenge of re-evaluating their business strategies in order to withstand the global crisis and save their successful business, which was suddenly at risk. In this case study, the owners presented their respective strategies to the students, who were then tasked with evaluating the pros and cons of each option based on the theories and concepts learned in class. This scenario provides students with the opportunity to apply their knowledge to a real-world problem and consider the potential implications of different courses of action.

3 METHODS

The participants in the pilot were business administration students taking part in a business strategy class. Their ages ranged between 23-40. The students were asked to experience the virtual scenario as part of the class curriculum, the same as any other required course material.

The virtual classroom at Reichman University (RU) features 60 standalone 4k 3DOF Pico G2 4K VR devices (Figure 1, B). To create immersive content that is relevant and specific to the school's curriculum, RU set up an immersive studio within the Innovation Center. This studio has been working in collaboration with the faculty to design and create immersive experiences that align with the school's educational goals. For example, students have had the opportunity to virtually meet with the owners of a Tel Aviv bar as they navigate strategic decisions related to surviving the COVID-19 pandemic. In addition to providing an immersive VR experience within the classroom, students were also given the option to engage with the content outdoors (Figure 1, C) or remotely via video conferencing (Figure 1, D). This flexibility allows for a more inclusive and accessible learning experience for all students.

The 360-degree video was integrated with interactive decision elements using the Unity software (Unity Inc.). The development of this immersive experience was based on the PICO SDK, and interactive buttons were implemented using gaze selection and button press on the main selection button of the PICO device (Figure 2). During each class, students were presented with 3-4 minutes of 360-degree content related to the case study and were asked to make decisions using the gaze and press function on the PICO device. These decisions were recorded live using Google Forms and displayed in real-time in the classroom via the Google Data Studio webapp¹ (Figure 2). Overall each student spent approximately 15 minutes in the virtual experience separated into 3 sections. The first section included an introduction to the entrepreneurs and the other sections included specific strategies presented by the entrepreneurs to contemplate on. This interactive approach allowed for a more engaging and relevant debate among students as they considered the different strategies presented throughout the case study.

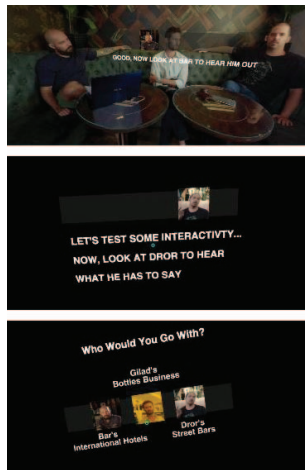


Figure 2: Interaction examples that were implemented within the case study experience

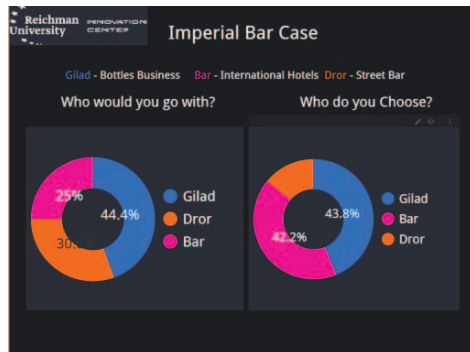


Figure 3: Representation of class dashboard results of both interaction questions used as part of the in-class discussion

4 RESULTS

We conducted a user evaluation of the immersive case study experience among 88 students. The evaluation included students who experienced the VR content on a 2D screen at home, a VR headset at home, a VR headset in the classroom with the lecturer, a VR headset in a Zoom classroom, and outdoors on campus while participating in a Zoom discussion. The overall evaluation of the

interactive experience was above the middle of the scale of the measure (2.5), with a mean score of 3.9 on a scale of 1-5. Most students reported high levels of information processing (mean of 3.6 out of 5), clarity in the class (mean of 3.81 out of 5), and pleasure and enjoyment (mean of 3.9 out of 5). These scores were above the median of 2.5, indicating high acceptance of the immersive case study experience among the students. Overall, our findings suggest that the immersive case study approach is well-received by students and can be effectively integrated into the classroom as a learning tool. Furthermore, our results suggest that by providing immersive content for students to access at home, we can create similar levels of engagement and attention as if they were physically present in the university classroom. This is a valuable conclusion as it suggests that immersive case studies can be an effective tool for delivering education remotely.

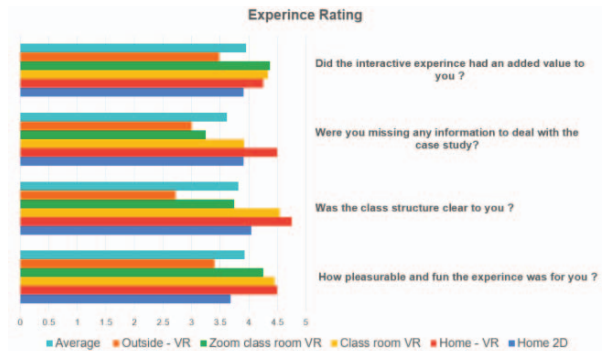


Figure 4: Experience rating results distributed according to viewing settings and average response (turquoise bar)

To date, we have collected data from approximately 800 participants in our immersive case study program. Of these participants, we have logged 421 decisions made within the VR application during class sessions.

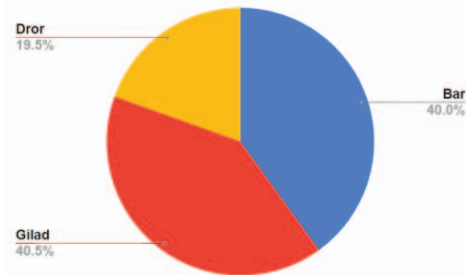


Figure 5: Aggregated strategy decisions from 421 students who participated in the immersive lessons.

5 CONCLUSIONS

This paper describes an innovative pilot program that has received, over the course of two years, prominent levels of acceptance among students, remaining a popular facet of the university's business administration program to this day. Our findings suggest that the use of immersive interactive technologies may be a viable and effective method for teaching social science material in higher education settings.

One of the major challenges in evaluating the use of immersive technologies in an educational setting is the issue of content adaptation and production. The cost of creating scenarios, whether through 360 video production or CGI projects, is often high. Additionally, the narrative and interactive elements of the

¹ <https://datastudio.withgoogle.com/>

experience need to be carefully developed in collaboration with the faculty members using the material.

During the project, we encountered several technological constraints. One such issue was the difficulty in controlling the class in sync using a class management tool. While we evaluated several potential solutions, none proved powerful enough, leading us to instruct students to operate the scene selection themselves, which proved effective. However, this approach requires an appointed worker to tutor students and assist professors, limiting the dissemination of the content.

Another constraint stems from the features of the chosen devices. We selected the PICO G2 4K, which at the time was the only enterprise device available in the quantities we needed (60 pieces). However, these devices only have 3DOF, which limits the level of immersion that can be achieved.

Maintaining the devices charged and updated is another issue that we faced during the project. A high-end class management (MDM) software could have helped to address this issue. However, the availability of such software was limited at the time of the project, making it difficult to find a suitable solution. It is important for educators considering the use of immersive technologies in their classrooms to carefully consider the technical infrastructure required to support such technologies, including the availability of suitable class management tools and the need to ensure that devices are charged and updated regularly.

6 FUTURE WORK

We are currently in the second stage of our project, intending to teach more than 50% of a dedicated course incorporated by immersive experiences for case study material dissemination. This means that in at least 50% of the classes in the selected course, the students will experience approximately 10 to 15 minutes of virtual content. To overcome the barriers associated with content creation, we have been exploring AI methods for content generation. Soon, we plan also to incorporate CGI-based education methods, allowing us to deliver 6DOF experiences with high levels of presence to our immersive experiences. These efforts will help to expand the experiences and effectiveness of immersive technologies in educational settings, allowing us to provide students with a more engaging and interactive learning experience.

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